Chat Log: Motor Driver Wiring Diagram

Conversation with Claude · exported October 28, 2026

YOU

hey, i need a wiring diagram for the following charts.

Table 1: Raspberry Pi 4 connections

Connected Component	Connected Pin / Node	RP4 Pin (Current)	Description
4N25 (via R1 330Ω)	Pin 1 (Anode)	GPIO24 (Physical 18)	H-Bridge Enable Signal
SN754410NE	Pin 2 (1A)	GPIO17 (Physical 11)	Motor Driver Logic 1
SN754410NE	Pin 7 (2A)	GPIO27 (Physical 13)	Motor Driver Logic 2
SN754410NE	Pin 10 (3A)	GPIO22 (Physical 15)	Motor Driver Logic 3
SN754410NE	Pin 15 (4A)	GPIO23 (Physical 16)	Motor Driver Logic 4
IR Sensor	OUT	GPIO13 (Physical 33)	High-Speed Polling Input
IR Sensor	VCC	3.3V (Physical 1)	Sensor Logic Power
Common GND Net	System Ground	GND (e.g., Physical 39)	Common Ground Reference

Table 2: 4N25 optocoupler

Connected Component	Connected Pin / Node	4N25 Pin (Current)	Description
RP4 (via R1 330Ω)	GPIO24 (Physical 18)	Pin 1	LED Anode
Common GND Net	System Ground	Pin 2	LED Cathode
(None)	Not Connected	Pin 3	NC
SN754410NE (& R2 10kΩ to GND)	Pin 1 (1,2EN) & Pin 9 (3,4EN)	Pin 4	Emitter (Enable Net)
External Power	+5V (VCC1)	Pin 5	Collector
(None)	Not Connected	Pin 6	Base (NC)

Table 3: SN754410NE H-bridge

Connected Component	Connected Pin / Node	SN754410NE Pin (Current)	Description
4N25 (& R2 10kΩ to GND)	Pin 4 (Emitter)	Pin 1 (1,2EN)	Enable 1,2
RP4 (& R3 1.8kΩ to GND)	GPIO17 (Physical 11)	Pin 2 (1A)	Logic Input 1
NEMA17 Motor	Green Wire	Pin 3 (1Y)	Motor Output 1
Common GND Net	System Ground	Pin 4 (GND)	Ground / Heat Sink
Common GND Net	System Ground	Pin 5 (GND)	Ground / Heat Sink
NEMA17 Motor	Grey Wire	Pin 6 (2Y)	Motor Output 2
RP4 (& R4 1.8kΩ to GND)	GPIO27 (Physical 13)	Pin 7 (2A)	Logic Input 2
External Power (& C2 100nF to GND)	+12V	Pin 8 (VCC2)	Motor Power Supply
4N25 (& R2 10kΩ to GND)	Pin 4 (Emitter)	Pin 9 (3,4EN)	Enable 3,4
RP4 (& R5 1.8kΩ to GND)	GPIO22 (Physical 15)	Pin 10 (3A)	Logic Input 3
NEMA17 Motor	Red Wire	Pin 11 (3Y)	Motor Output 3
Common GND Net	System Ground	Pin 12 (GND)	Ground / Heat Sink
Common GND Net	System Ground	Pin 13 (GND)	Ground / Heat Sink
NEMA17 Motor	Yellow Wire	Pin 14 (4Y)	Motor Output 4
RP4 (& R6 1.8kΩ to GND)	GPIO23 (Physical 16)	Pin 15 (4A)	Logic Input 4
External Power (& C1 100nF to GND)	+5V	Pin 16 (VCC1)	Logic Power Supply

Table 4: NEMA17 motor

Connected Component	Connected Pin / Node	Motor Wire (Current)	Description
SN754410NE	Pin 3 (1Y)	Green	Coil 1 (~37 ohm pair)
SN754410NE	Pin 6 (2Y)	Grey	Coil 1 (~37 ohm pair)
SN754410NE	Pin 11 (3Y)	Red	Coil 2 (~36.3 ohm pair)
SN754410NE	Pin 14 (4Y)	Yellow	Coil 2 (~36.3 ohm pair)

Table 5: IR sensor

Connected Component	Connected Pin / Node	IR Sensor Pin (Current)	Description
RP4	3.3V (Physical 1)	VCC	3.3V Logic Power
Common GND Net	System Ground	GND	Ground
RP4	GPIO13 (Physical 33)	OUT	Digital Edge Output

CLAUDE

Your tables describe four separate groups of connections, so I split the wiring into three diagrams to keep each one readable.

[Diagram 1: the 4N25 opto-gated enable path. Diagram 2: the four logic lines from the Pi to the driver, and the driver outputs to the NEMA17 coils. Diagram 3: the driver's power and ground pins with bypass caps, and the IR sensor wiring. These were interactive visuals in the chat and are superseded by the combined diagram later in this document.]

A few things to check before you power it up:

- **Common ground:** All three grounds (Pi, 5V/12V supplies, driver) are tied together, as your tables specify. That's required for the logic signals to work, but it means the 4N25 isn't giving you galvanic isolation. It acts as a switched enable gate only.
- **LED current:** R1 at 330Ω gives roughly (3.3V − 1.2V) / 330Ω ≈ 6 mA from GPIO24. That's fine for the Pi and enough to saturate the 4N25 output into a 10kΩ load.
- **Motor current:** Each coil is about 36 to 37Ω, so 12V gives roughly 0.32 A per winding. That's well within the SN754410's 1 A per channel rating.
- **Coil pairing:** Green/Grey are one coil and Red/Yellow the other. If the motor jitters instead of turning, swap one pair's wires.
- **Pulldowns:** The 1.8kΩ resistors hold the driver inputs low while the Pi boots, since GPIOs float until configured.

I can also turn this into a single combined schematic, or a pinout table with wire colors for building it on a breadboard, if you'd like.

YOU

can you make one diagram, looping all the components together?

CLAUDE

Here's everything in one diagram. All the GND stubs go to the same common ground net.

[Combined diagram: see the final version below, which includes the changes from the next request.]

This is the complete circuit on one page, with the same connections as your tables.

- **Enable bus:** The 4N25 emitter feeds pins 1 and 9 of the driver through the horizontal bus above the SN754410NE, with R2 pulling it to ground.
- **GND stubs:** Every GND label is the same net: the Pi, both supplies, the 4N25 cathode, R2 through R6, the capacitors, the driver's four GND pins, and the IR sensor.

- **Capacitors:** C1 (100nF on the 5V pin) and C2 (100nF on the 12V pin) should sit as close to the driver's pins as you can get them.

I can also export this as a standalone SVG or PDF, or make a printable pin-by-pin wire list for assembly if you want one.

YOU

can you label the enable pins on the HBridge and show the capacitor connections as well for the power inputs?

CLAUDE

I've added the enable pin labels inside the H-bridge, and the two bypass capacitors now appear as drawn symbols on the 5V and 12V inputs.

The updated diagram has these changes:

- **Enable pins:** The top of the SN754410NE now shows pin 1 (1,2EN) and pin 9 (3,4EN), both fed from the 4N25 emitter bus. The chip name moved to the middle of the box to make room.
- **C1:** 100nF from the 5V line at VCC1 (pin 16) to ground.
- **C2:** 100nF from the 12V line at VCC2 (pin 8) to ground.

Each capacitor taps its supply line right at the driver pin, which is where bypass caps work best. If you use an electrolytic bulk cap on the 12V motor supply as well (something like 100µF), it would go in parallel with C2.

[Final combined diagram: see next page.]

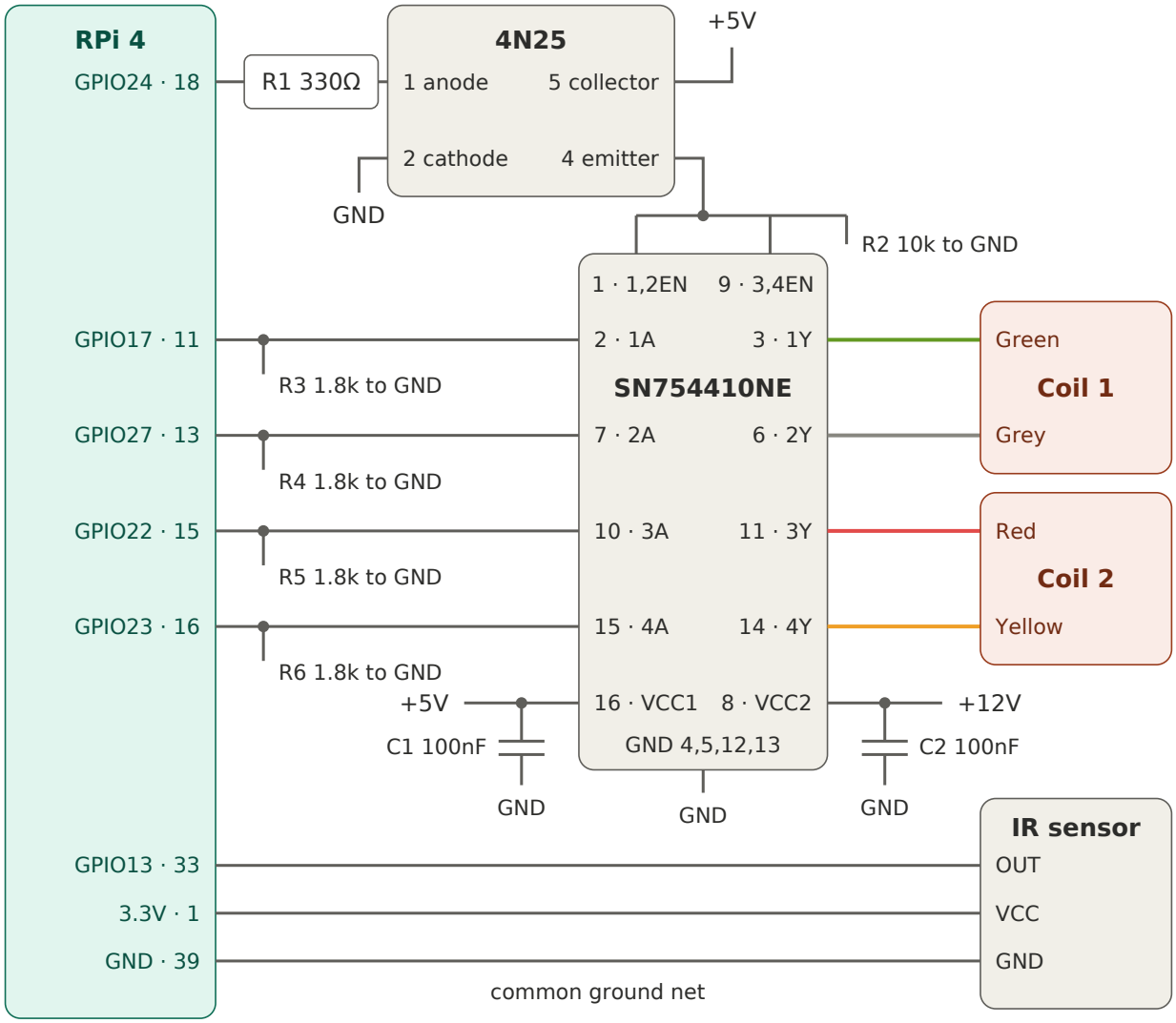
YOU

could make me an exportable copy of our chat logs as a pdf?

CLAUDE

This document.

Final combined wiring diagram



Wire colors on the motor outputs match the motor lead colors (Green, Grey, Red, Yellow). All GND labels are one common ground net. Redrawn from the final diagram in the chat.